



ABES Engineering College, Ghaziabad
(An Autonomous Institute)

Department of Applied Sciences & Humanities

Session: 2025-26

Semester: II

Course Code: 25AS202

Course B.Tech

Name: Applied Calculus & Differential Equations

Assignment 2

Date of Assignment:

Date of submission:

S.No.	KL	CO	Question	Marks
1	K3	CO1	Verify that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$, where $u(x, y) = \log \left(\frac{x^2 + y^2}{xy} \right)$	
2.	K3	CO1	If $u = x^2 \tan^{-1} \left(\frac{y}{x} \right) - y^2 \tan^{-1} \left(\frac{x}{y} \right)$, Find the value $\frac{\partial^2 u}{\partial x \partial y}$. Ans: $\frac{x^2 - y^2}{x^2 + y^2}$	
3.	K3	CO1	If $u = \sin^{-1} \left(\frac{x^3 + y^3 + z^3}{ax + by + cz} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 2 \tan u$.	
4	K3	CO1	If $u = x f \left(\frac{y}{x} \right) + \phi \left(\frac{y}{x} \right)$, show that (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = x f \left(\frac{y}{x} \right)$ (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0$	
5	K3	CO1	If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$, show that (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \cos 3u \sin u$	
6	K3	CO1	If $u = x^4 y^2 \sin^{-1} \left(\frac{x}{y} \right) + \log x - \log y$, show that (i) $x u_x + y u_y = 6x^4 y^2 \sin^{-1} \left(\frac{x}{y} \right)$. (ii) $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 30u$	

7	K3	CO1	If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, prove that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0, \text{ i.e. } 6u_x + 4u_y + 3u_z = 0$	
8	K3	CO1	If u, v, w are the roots of the cubic $(x-a)^3 + (x-b)^3 + (x-c)^3 = 0 \text{ then find } \frac{\partial(u, v, w)}{\partial(a, b, c)}.$ Ans: $-\left[\frac{2(a-b)(b-c)(c-a)}{(u-v)(v-w)(w-u)} \right]$	
9	K3	CO1	If $u = x + y + z$, $v = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx$ and $w = x^3 + y^3 + z^3 - 3xyz$ then show that they are not independent. Find the relation between them. Ans: $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$; $u(u^2 + 3v) = 4w$.	
10	K3	CO1	Find the stationary points of $f(x, y) = x^4 + y^4 - 2(x - y)^2$ and determine their nature. Also find the extreme value of function. Ans. Max. at (0,0), max. value=0;	